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Experiment (2) Report

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Faculty of Engineering - Ain Shams University
Electronics and Electrical Communications Engineering Department

ECE352s Analog and Digital Communication Systems

Junior - Electronics and Electrical Communications

Experiment (2)

Analog Modulation

Spring 2026

1. Objectives:

- To be familiar with different analog modulation techniques.
- Send and receive single tone signal using analog modulation schemes.

2. Equipment:

- Digital CRO
- Telecoms-Trainer 101 KIT

3. Part 1: Amplitude Modulation and Demodulation

In an amplitude modulation (AM) communications system, the message or baseband signal is used to electrically vary the amplitude of a pure sinewave called the carrier. The carrier usually has a frequency that is much higher than the message's frequency.

Figure 1 below shows a simple message signal and an unmodulated carrier. It also shows the result of amplitude modulating the carrier with the message. Notice that the modulated carrier's amplitude varies above and below its unmodulated amplitude.

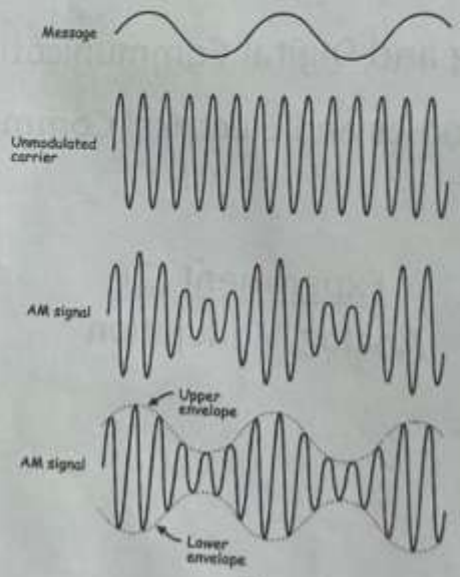


Figure 1

In telecommunications theory, the mathematical model that defines the AM signal is:

$$\text{AM} = (\text{DC} + \text{message}) \times \text{the carrier}$$

In this experiment you'll generate a real AM signal by implementing its mathematical model, then you will setup an envelope detector using the rectifier and RC LPF, you'll vary the message signal's amplitude and observe how it affects the modulated carrier. You'll observe the effects of modulating the carrier too much, and measure the AM signal's depth of modulation using a scope.

Procedure

Part A - Generating an AM signal using a simple message

1. Set the scope's Channel 1 Input Coupling control to the DC position.
2. Locate the Adder module and turn its G and g controls fully anti-clockwise.
3. Locate the Variable DCV module and turn its DC Voltage control almost fully anti-clockwise.
4. Connect the set-up shown in Figure 2.
   At the moment, the scope should just be showing a flat trace because the Adder module's output is 0V.
7. Use the scope's Channel 1 Vertical Position control to move the trace so that it lines up with the horizontal line in the middle of the scope's screen.
8. While watching the Adder module's output on the scope, turn its g control clockwise until the DC level is 1V, then turn its G control clockwise to obtain a $1V_{p-p}$ sinewave.
12. With the setup you already have (dotted in figure 3) put the highlighted wires (in black), with these additions the AM equation should be: AM= (1+$1V_{p-p}$ 2kHz sine) × $4V_{p-p}$ 100kHz sine
13. Use the scope's CH1 vertical position to overlay the message with the AM envelope and compare them.
14. Draw the two waveforms to scale.

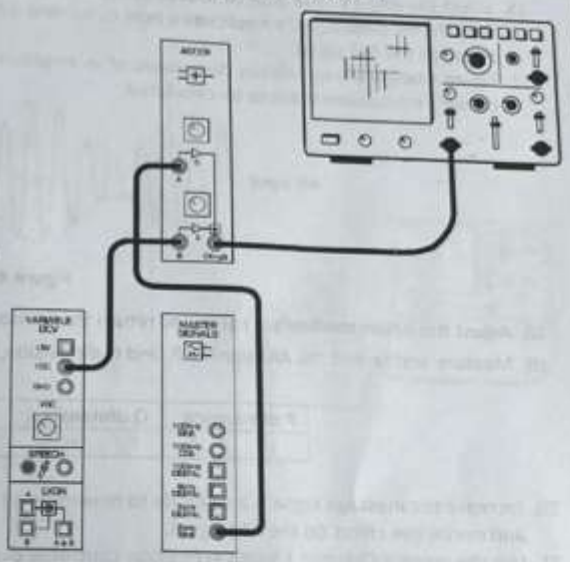


Figure 2

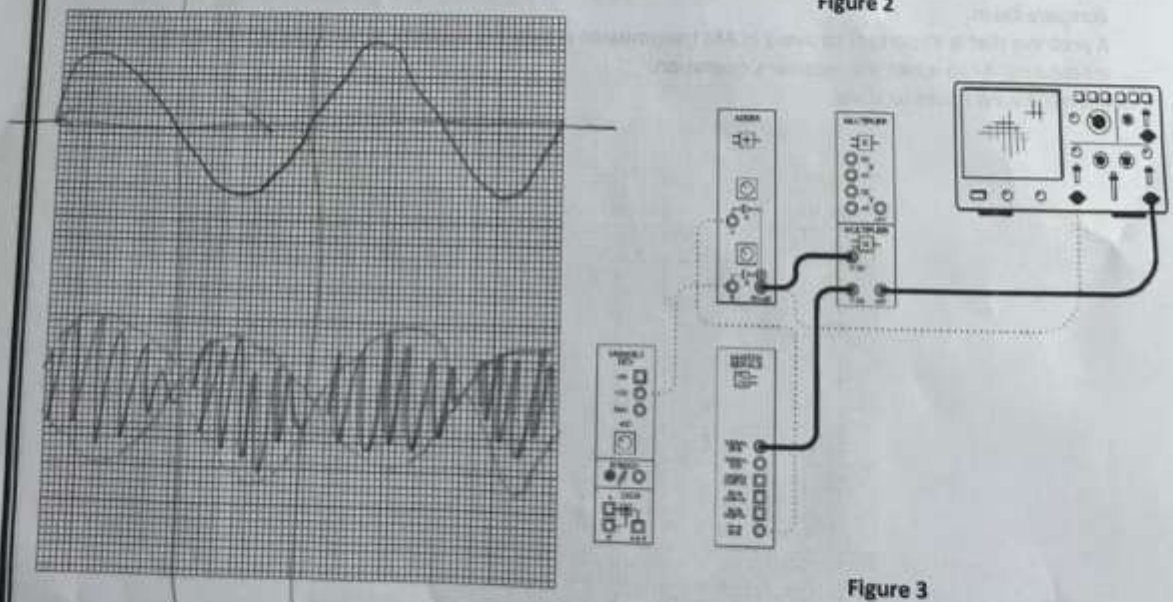


Figure 3

It's possible to modulate the carrier by different amounts.
The extent that a message modulates a carrier is known in the industry as the modulation index (m).
Modulation index is an important characteristic of an AM signal for several reasons including calculating the distribution of the signal's power between the carrier and sidebands.

15. *adjust the scope's Time base control to be 0.1ms/div position.*
16. *Vary the message signal's amplitude a little by turning Adder module's G control left and right and notice the effect on the AM signal.*
17. *Figure 4 below shows two key dimensions of an amplitude modulated carrier. These two dimensions allow a carrier's modulation index to be calculated.*

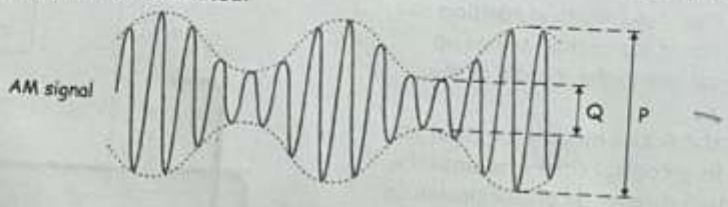
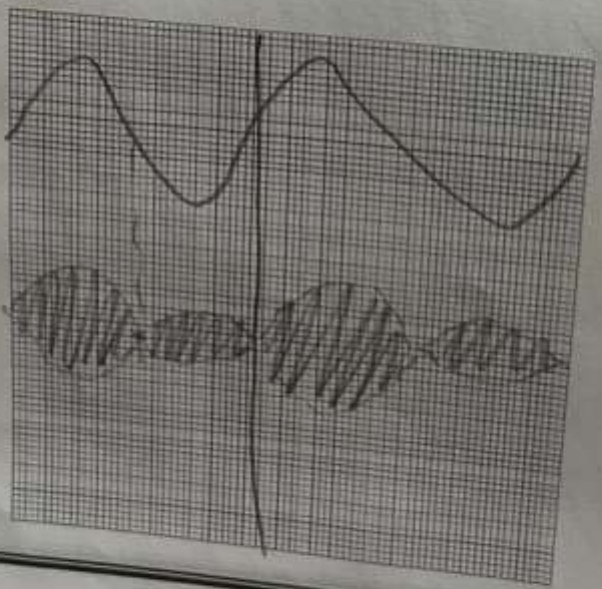


Figure 4

18. Adjust the Adder module's G control to return the message signal's amplitude to $1V_{p-p}$.
19. Measure and record the AM signal's P, and Q dimension, hence calculate $m = \frac{P-Q}{P+Q}$ Record your measurement.

P dimension	Q dimension	Modulation index (m)
5.68	2.08	0.464

20. Increase the message signal's amplitude to maximum by turning the Adder module's G control fully clockwise and notice the effect on the AM signal.
21. Use the scope's Channel 1 Vertical Position control to overlay the message with the AM signal's envelopes and compare them.
    A problem that is important to avoid in AM transmission is over-modulation. When the carrier is over-modulated, it can upset the receiver's operation.
22. Draw the waveforms to scale.



Part B - Receiving AM signal using envelope detector (non-coherent detection)

Recovering the original message from a modulated carrier is called Demodulation and this is the main purpose of communications and telecommunications receivers. The circuit that is widely used to demodulate AM signals is called an envelope detector. The block diagram of an envelope detector is shown in Figure 5

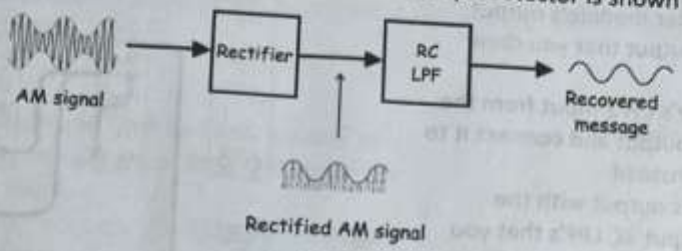


Figure 5

1. Modify the set-up as shown in Figure 6.
2. Adjust the scope's Vertical Attenuation controls to appropriate settings for the signals.
3. Draw the two waveforms to scale in the space provided below leaving room to draw a third waveform.
4. Disconnect the scope's Channel 2 input from the Rectifier's output and connect it to the RC LPF's output instead.
5. Draw the demodulated AM signal to scale in the space, and compare the three waveforms.
6. Vary the message signal's amplitude up and down a little (by turning the Adder module's G control left and right a little) while watching the demodulated signal. Comment.

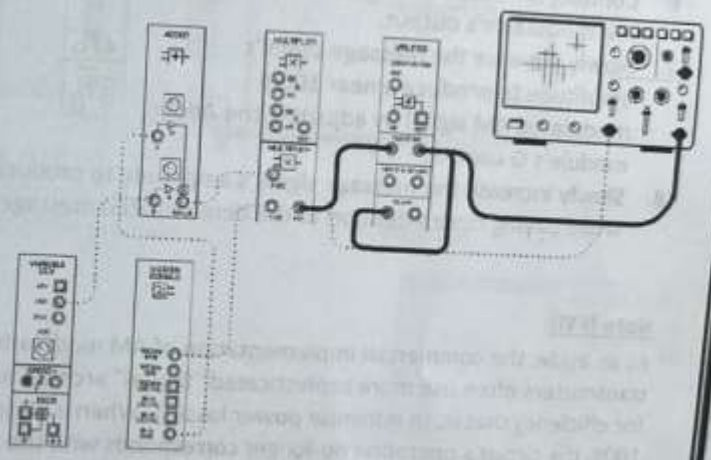


Figure 6

The msg can't be received.

7. Slowly increase the message signal's amplitude to maximum while watching the demodulated signal. Comment.

phase reverse.

Part C - Receiving AM signal using product demodulation (coherent detection)

1. Modify the setup to return it as an AM modulator with $1V_{p-p}$ 2KHz sine wave message signal.
2. Now connect the setup as shown in figure 7.
3. Compare the Multiplier module's output with the Rectifier's output that you drew earlier.
4. Disconnect the scope's CH 2 input from the Multiplier module's output and connect it to the RC LPF's output instead.
5. Compare the RC LPF's output with the message and the output RC LPF's that you drew earlier.
6. Connect the scope's Channel 1 input to the AM modulator's output.
7. Slowly increase the message signal's amplitude to produce a near 100% modulated AM signal by adjusting the Adder module's G control.
8. Slowly increase the message signal's amplitude to produce an AM signal that is modulated by more than 100% while paying close attention to the demodulated message signal. Comment.

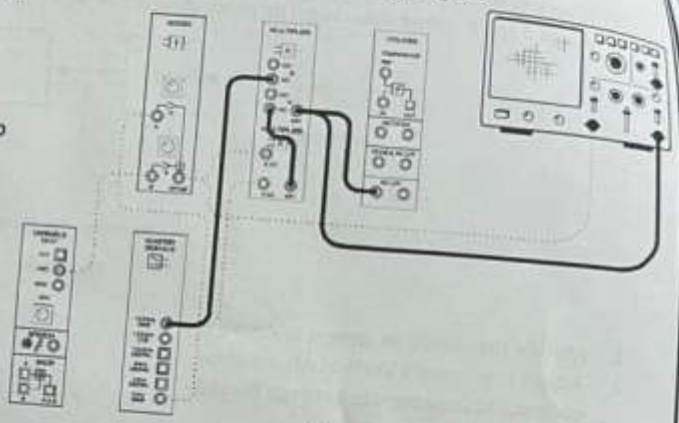


Figure 7

Note (FYI):

As an aside, the commercial implementation of AM modulation commonly involves a Class C amplifier (modern transmitters often use more sophisticated "Digital" architectures like Class D or E switching or Pulse Step Modulation for efficiency that is, to minimise power losses). When a Class C amplifier is operated at depths of modulation above 100% the circuit's operation no-longer corresponds with the model of an AM modulator. Importantly, in addition to producing an envelope that is not the same as the original message, the over-modulated Class C circuit produces extra frequency components in the spectrum. This means that neither the envelope detector nor the product demodulator can reproduce the message without distortion.

4. Part 2: Frequency Modulation and Demodulation

A disadvantage of the AM communication systems is that they are susceptible to picking up electrical noise in the transmission medium (the channel). This because noise changes the amplitude of the transmitted signal and the demodulators of these systems are affected by amplitude variations.

Frequency modulation (FM) uses a message's amplitude to vary the frequency of a carrier instead of its amplitude. This means that the FM demodulator is designed to look for changes in frequency instead. As such, it is less affected by amplitude variations and so FM is less susceptible to noise. This makes FM a better communications system in this regard.

Figure 8 shows a simple message signal (a bipolar square wave) and an unmodulated carrier. It also shows the result of frequency modulating the carrier with the message.

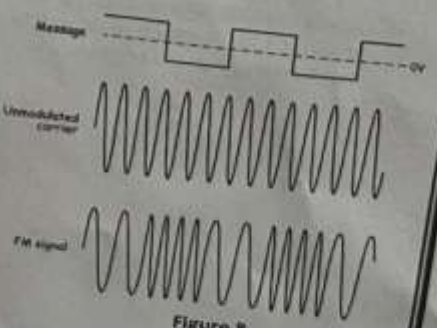


Figure 8

Procedure

Part A - Generating an FM signal using a simple message

You'll generate a real FM signal using the VCO module on the Emona Telecoms-Trainer 101. First, you'll set up the VCO module to output an unmodulated carrier at a known frequency. Then you'll observe the effect of frequency modulating its output with a square wave. Finally, you'll use a sinewave to observe the spectral composition of an FM signal (in the time domain).

1. Locate the VCO module and turn its Gain control to about two thirds of its travel (about the position of the number 2 on a clock face).
2. Set the VCO module's Frequency Adjust control to about the middle of its travel.
3. Set the VCO module's Range control to the LO position.
4. Connect the set-up shown in Figure 9.
5. Set the scope's Time base control to the 20µs/div position.
6. Adjust the VCO module's Frequency Adjust control so that one cycle of its output is exactly 5 divisions.
7. Set the scope's Time base control to the 0.1ms/div position.
8. Modify the set-up as shown in Figure 10.
9. The setup in Figure 10 can be represented by the block diagram below. The Master Signals module is used to provide a 2kHz square wave message signal and the VCO module is the FM modulator with a 10kHz carrier.
10. If necessary, tweak the VCO module's Gain control until you obtain an output from the VCO that's similar to the FM signal in Figure 8.
11. Use the scope's Channel 1 Vertical Position control to overlay the message with the FM signal and compare them.

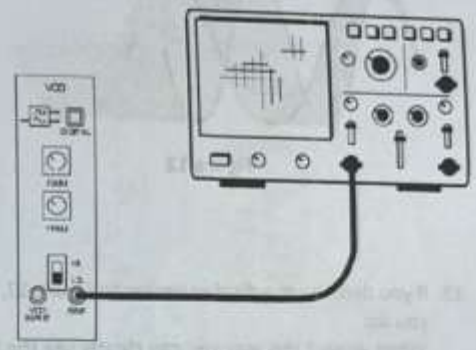


Figure 9

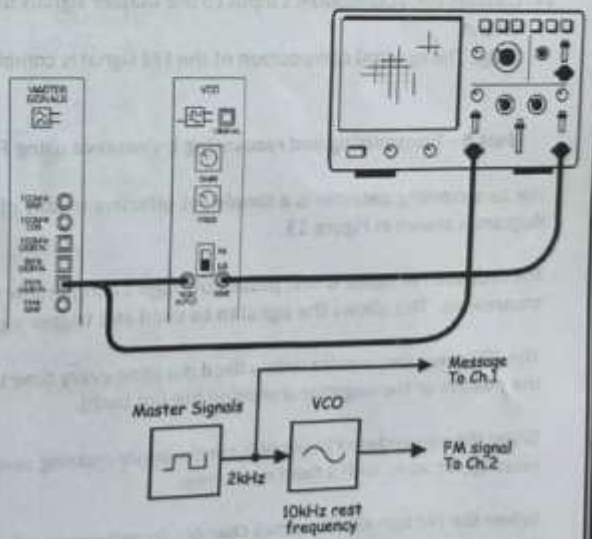


Figure 10

12. **Considering the spectral composition of FM signals** Regardless of the type of message signal used the spectral composition of FM signals is rich in sinewaves.

Modify the setup as shown in Figure 11. You should now see a display that looks similar to Figure 12.

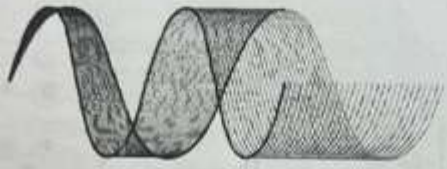


Figure 12

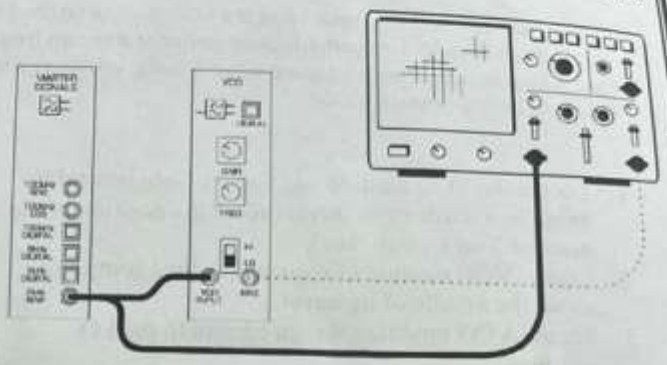


Figure 11

13. If you don't have a display similar to Figure 12, slowly turn the VCO module's Gain control anti-clockwise until you do.
    When viewed this way you can clearly see the highest frequency sinewave that the FM modulator is outputting, the lowest frequency sinewave and many of the sinewaves in between.
14. Connect the VCO module's input to the Master Signals module's 2kHz DIGITAL output instead of the 2kHz SINE output.
    Note: The spectral composition of the FM signal is complex regardless of the message's waveshape.

Part B - Transmitting and recovering a sinewave using FM

The zero-crossing detector is a simple yet effective means of recovering the message from FM signals. Its block diagram is shown in Figure 13.

The received FM signal is first passed through a comparator to heavily clip it, effectively converting it to a squarewave. This allows the signal to be used as a trigger signal for the zero-crossing detector circuit (ZCD).

The ZCD generates a pulse with a fixed duration every time the squared-up FM signal crosses zero volts (either on the positive or the negative transition but not both).

Given the squared-up FM signal is continuously crossing zero, the ZCD effectively converts the squarewave to a rectangular wave with a fixed mark time.

When the FM signal's frequency changes (in response to the message), so does the rectangular wave's frequency. Importantly though, as the rectangular wave's mark is fixed, changing its frequency is achieved by changing the duration of the space and hence the signal's mark/space ratio (or duty cycle).

Importantly, this demodulation technique works equally well when the message is a sinewave or speech.

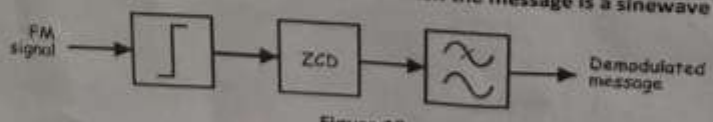


Figure 13

This part of the experiment lets you use the setup to modulate, transmit and demodulate a test signal (a sine wave).

15. Modify the setup as shown in Figure 14
16. Set the scope's Channel 2 Input Coupling control to the AC position.
17. Adjust the scope's Time base control to view two or so cycles of the Master Signals module's 2kHz SINE output.
18. Compare the message with the FM demodulator's output.
    Note: If your set-up is working correctly, the FM demodulator's output should be the same as the message (with some phase shift).

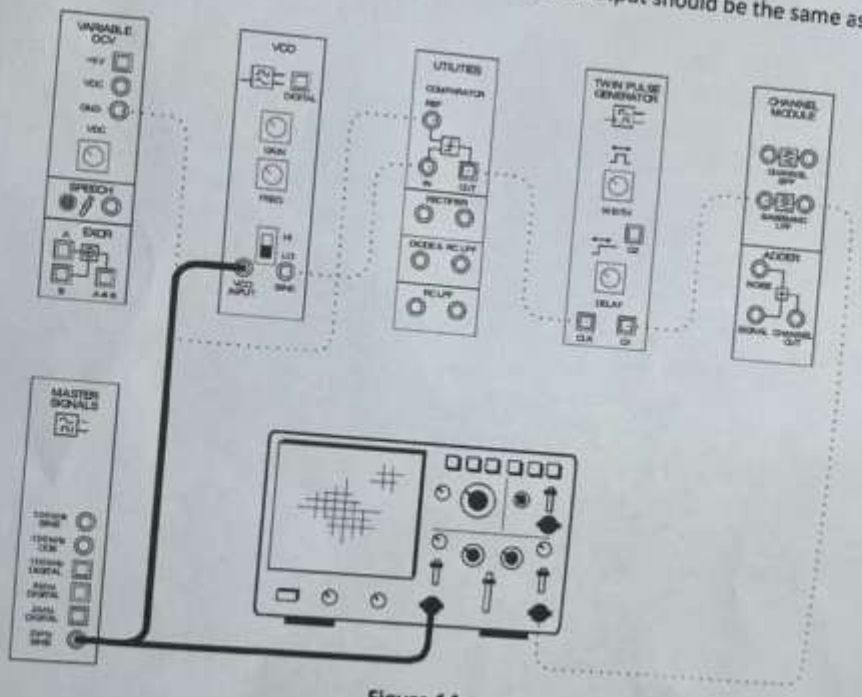
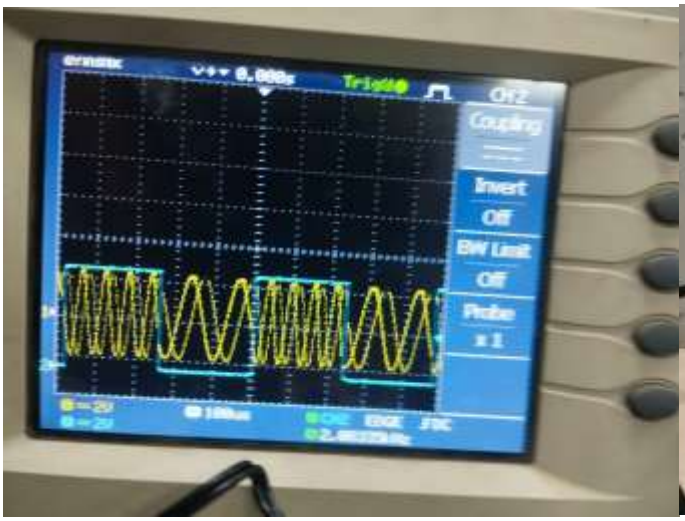
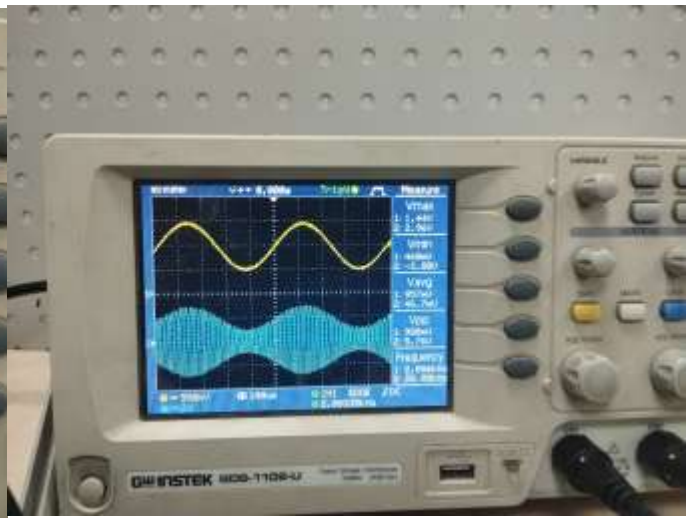


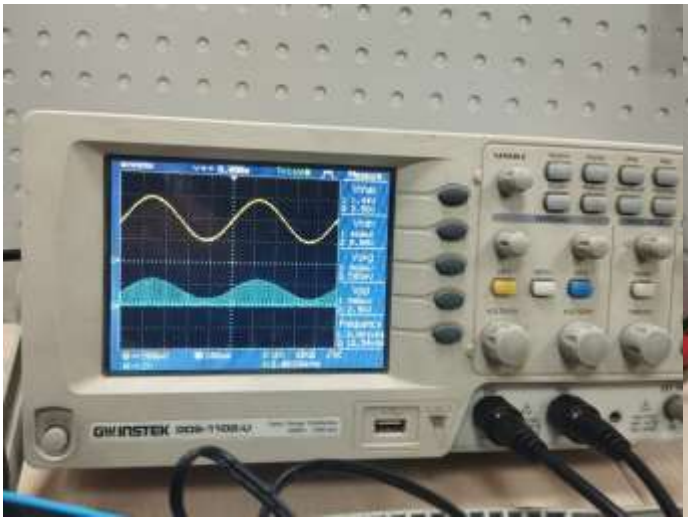
Figure 14

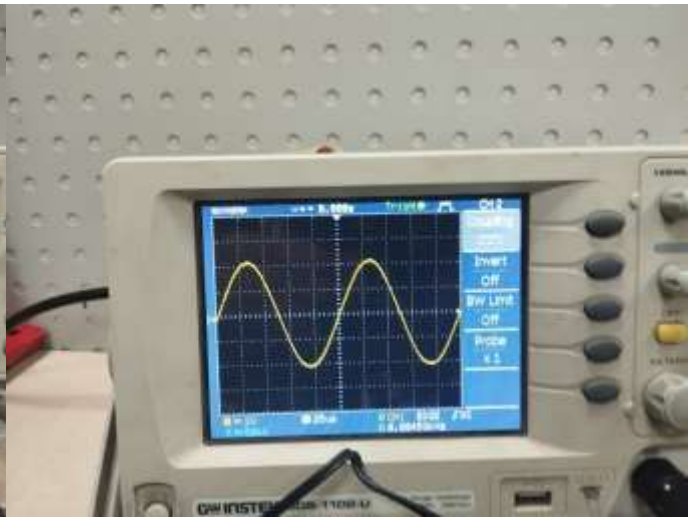
19. What does the FM modulator's output signal tell you about the ZCD signal's duty cycle?
    Hint: Use the scope's Channel 2 input to examine the output of the ZCD.

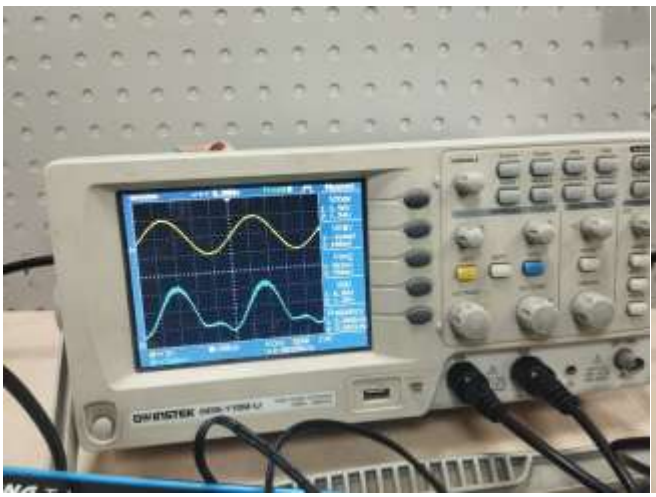
Pictures of output waveforms on oscilloscope

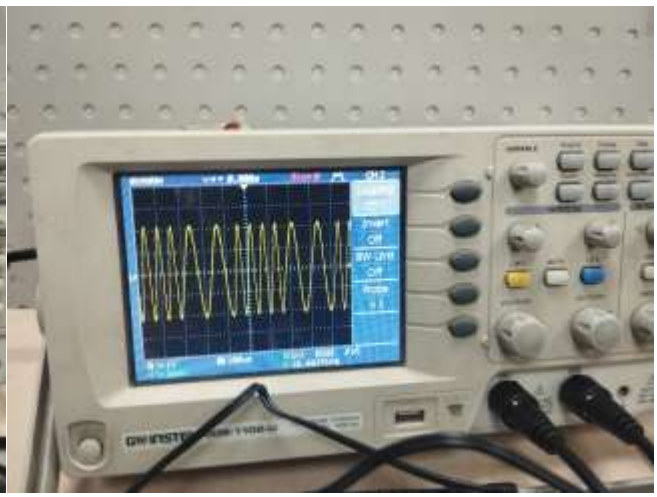


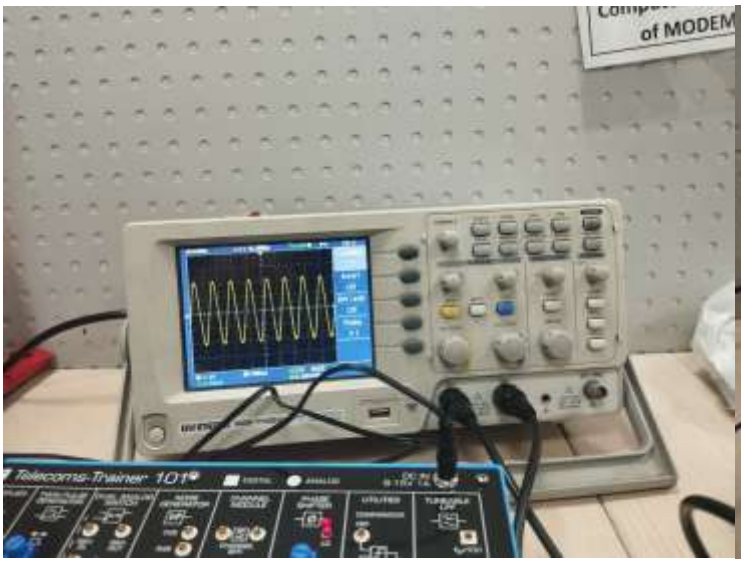











of MODEM
Telecoms-Trainer 101

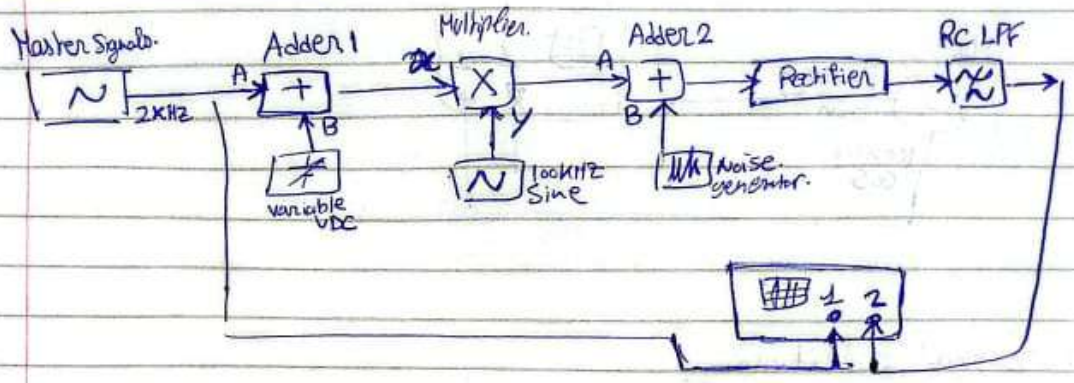



Computer Simulation
of MODEMs
HONA Telecoms-Trainer 101

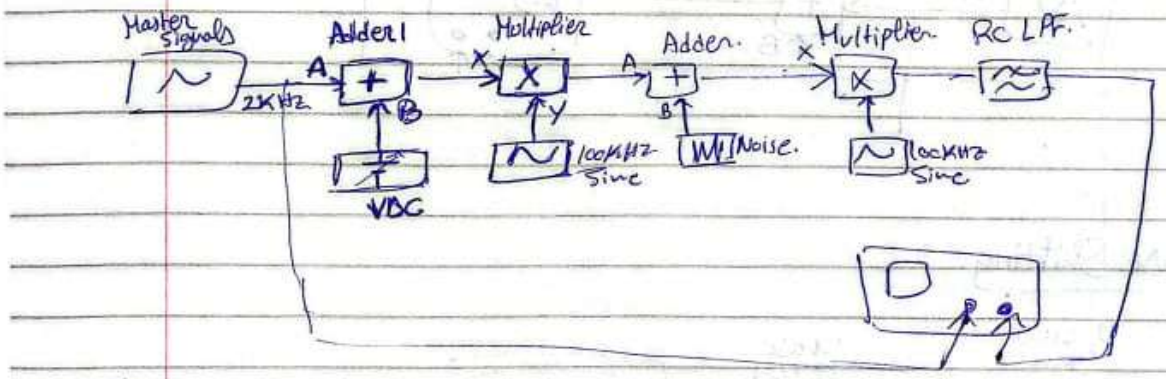

Computer Simulation
of MODEMs
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ADC Report 2
Block diagrams

1) AM mod. with envelope detection



2) AM with product detection



3) FM

